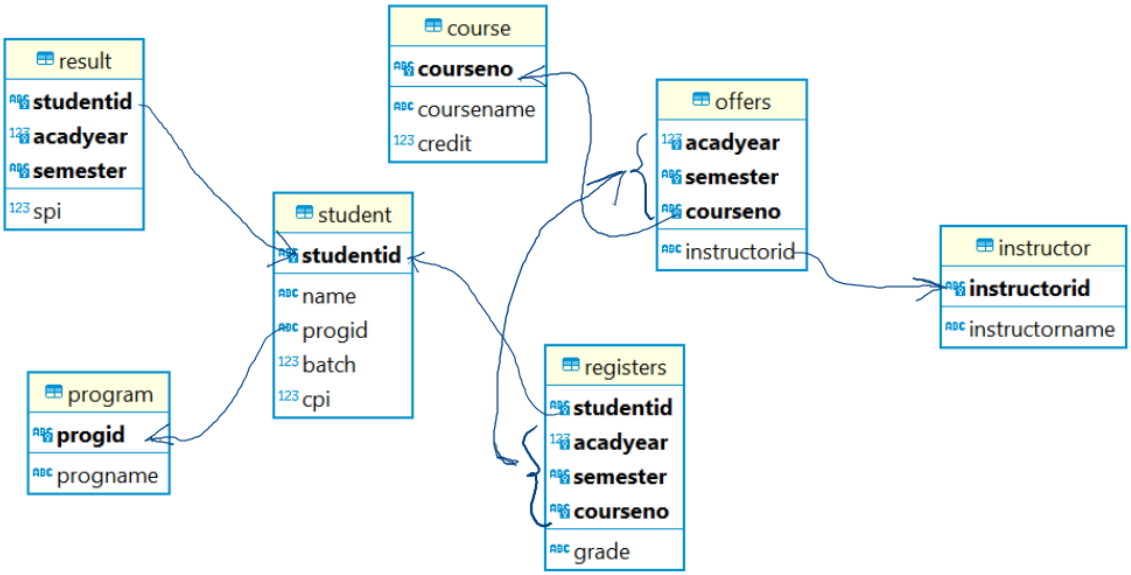
Lab04	Querying operations on relations .2
DS604 Introduction to Data Management, Autumn'2026; pm_jat @ daiict	

In this lab too, you practice querying databases using Relational Algebra and SQL.

Here you write and perform some queries on the **ACAD** database. The database instance (named as **acad**) with some sample data is also available in the **"public" database** on the same server.

Here is database schema diagram for your reference:



Your goal here is to write relational algebra expressions and SQL statements for answering the following queries. [Algebraic expressions can be handwritten and scanned for submission]

1. List Faculty (ID, Name) for all courses of Autumn'2010
2. List Students (ID, Name) from batch 2008 registered for course MT101 in Autumn'2008.
3. List Students (ID, Name) from B Tech (progid='01') batch 2008 registered for course MT101 in Autumn'2008.
4. List courses that were offered but students were not registered in those offerings
5. List students (id, name, courseno, grade) who got 'F' grade in Autumn'2008
6. List Courses (Course No, Title, Credits) offered by Instructor (ID='PMJ') in semester Winter'2010.
7. Produce transcript of student '200711002' for Academic year 2008-09
   List Course No, Course Name, Course Credit, Grade
8. List ID of students who have taken both courses - 'MT101' AND 'MT104'.
9. List ID of students who have taken 'MT101' but not 'MT104'

[Submission Required: Query Expressions in Relational Algebra and SQL]

Submission Instructions: Submit everything in a single PDF file.